



WISCONSIN
UNIVERSITY OF WISCONSIN-MADISON

Name: Solutions

Math 421 - Section 2 - Exam 1

Fall Semester 2024

October 10, 2024 - 1:00 pm to 2:15 pm

Instructions:

Prob.	Points possible	Score
1	6	
2	4	
3	10	
4	10	
5	10	
6	10	
TOTAL	50	

- Read each problem carefully.
- Write legibly.
- Show all your work on these sheets. Feel free to use the opposite side.
- This exam has 6 pages, and 5 problems. Please make sure that all pages are included.
- You may not use books, notes, calculators, etc.

Good luck!

Question 1 (TRUE or FALSE). (6 points, 1 points each) Determine if each statement is true or false. You do not need to justify your answer.

1. F The statement "If P , then Q " is logically equivalent to the statement "If (not P), then (not Q)."
2. F The statement "not(P and Q)" is logically equivalent to the statement "(not P) and (not Q)."
3. F If a is irrational and b is rational, then $a \cdot b$ is irrational. ($b = 0$)
4. F If a is irrational and b is irrational, then $a + b$ is irrational. ($b = -a$)
5. T If (a_n) converges, then (a_n) is bounded.
6. T If (a_n) is bounded and monotone, then (a_n) converges.

Question 2. (4 points) Determine the minimum, maximum, infimum, and supremum of the following set. You do not need to justify your answer.

$$A = \left\{ 1 - \frac{1}{n} : n \in \mathbb{N} \right\}.$$

Minimum: 0

Maximum: D.N.E

Infimum: 0

Supremum: 1

Question 3. (Total of 10 points) Prove: If a is rational and b is irrational, then $a + b$ is irrational.

We argue by contradiction. Suppose $a + b$ is rational. Then $a = \frac{p_1}{q_1}$ and $a + b = \frac{p_2}{q_2}$ for some $p_1, p_2, q_1, q_2 \in \mathbb{Z}$ with q_1, q_2 nonzero.

Then

$$b = a + b - a = \frac{p_2}{q_2} - \frac{p_1}{q_1} = \frac{q_1 p_2 - p_1 q_2}{q_2 q_1}.$$

Since $q_1 p_2 - p_1 q_2 \in \mathbb{Z}$ and $q_2 q_1 \in \mathbb{Z}$,

b is rational. Contradiction!

Question 4. (Total of 10 points) Let (a_n) be the sequence where $a_1 = 1$, $a_2 = 2$, and

$$a_n = a_{n-1} + 3a_{n-2} \text{ when } n \geq 3.$$

Prove that $a_n \leq 3^n$ for all $n \geq 1$.

We argue by complete/strong induction.

Base Case: If $n=1$, then

$$a_1 = 1 < 3 = 3^1.$$

So true when $n=1$.

Induction Step Suppose $k \in \mathbb{N}$ and $a_m \leq 3^m$ for all $1 \leq m \leq k$.

Case 1: Assume $k=1$. Then

$$a_{k+1} = a_2 = 2 < 9 = 3^2.$$

So true for $k+1$.

Case 2: Assume $k \geq 2$. Then $k+1 \geq 3$. So

$$\begin{aligned} a_{k+1} &= a_k + 3a_{k-1} \leq 3^k + 3 \cdot 3^{k-1} = 3^k + 3^k \\ &= 2 \cdot 3^k < 3 \cdot 3^k = 3^{k+1}. \end{aligned}$$

So true for $k+1$.

Question 5. (Total of 10 points) Prove, using only the definition, that $\lim_{n \rightarrow \infty} \frac{n}{4n+1} = \frac{1}{4}$.

Fix $\epsilon > 0$. Let $N \in \mathbb{N}$ satisfy
 $N > \frac{1}{\epsilon}$.

Assume $n > N$. Then

$$\begin{aligned} \left| \frac{n}{4n+1} - \frac{1}{4} \right| &= \left| \frac{4n - (4n+1)}{(4n+1)4} \right| = \left| \frac{-1}{16n+4} \right| \\ &= \frac{1}{16n+4} < \frac{1}{16n} < \frac{1}{n} < \frac{1}{N} < \frac{1}{\epsilon} = \epsilon. \end{aligned}$$

Since $\epsilon > 0$ was arbitrary, $\lim_{n \rightarrow \infty} \frac{n}{4n+1} = \frac{1}{4}$.

Question 6. (Total of 10 points) Prove, using only the definition, that: if (a_n) is a sequence where $a_n \geq 0$ for all n and $\lim a_n = a$, then

$$\lim \frac{1}{7+a_n} = \frac{1}{7+a}.$$

Fix $\epsilon > 0$. Since $\lim a_n = a$, there exists $N \in \mathbb{N}$ such that: if $n \geq N$, then $|a_n - a| < \min\{\epsilon, 1\}$.

Assume $n \geq N$. Then

$$|a_n - a| < 1 \Rightarrow a - 1 < a_n < a + 1 \Rightarrow a_n - 1 < a.$$

Since $a_n \geq 0$, we have $a > -1$.

Then,

$$\begin{aligned} \left| \frac{1}{7+a_n} - \frac{1}{7+a} \right| &= \left| \frac{(7+a) - (7+a_n)}{(7+a_n)(7+a)} \right| = \left| \frac{a - a_n}{(7+a_n)(7+a)} \right| \\ &= \frac{|a_n - a|}{|7+a_n||7+a|}. \end{aligned}$$

Since $a_n \geq 0$, $|7+a_n| \geq 7$. Since $a > -1$, $|7+a| \geq 6$.

So

$$\left| \frac{1}{7+a_n} - \frac{1}{7+a} \right| \leq \frac{|a_n - a|}{7 \cdot 6} \leq |a_n - a| < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim \frac{1}{7+a_n} = \frac{1}{7+a}$.